SKILL ENHANCEMENT COURSE (SEC) PROJECT

Design and Development of an Intelligent USB-Based Electronics Platform

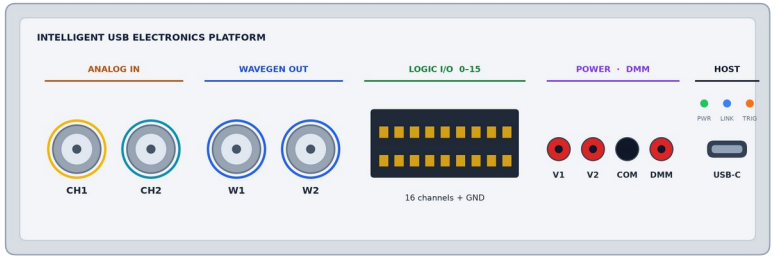
Eight bench instruments on one USB cable, with Proto, an AI engineer that plans every test and explains every measurement.

PROJECT TEAM

Avi Prakash Jaiswal	24294917
Soham Pandit Nalawade	24294917002
Harshit Saini	24294917086
Anshul	24294917037

SUPERVISOR

Prof. G.S. Chilana



ONE DEVICE, EIGHT INSTRUMENTS

- **Oscilloscope**
- **Waveform generator**
- **Logic analyser**
- **Pattern generator**
- **Protocol decoder**
- **Power supply 1**
- **Power supply 2**
- **Multimeter**

+ Proto
an AI engineer that runs the tests and gives a verdict in plain words

MOTIVATION AND OBJECTIVES

Why one intelligent platform instead of a bench of instruments

THE PROBLEM

Costly and bulky

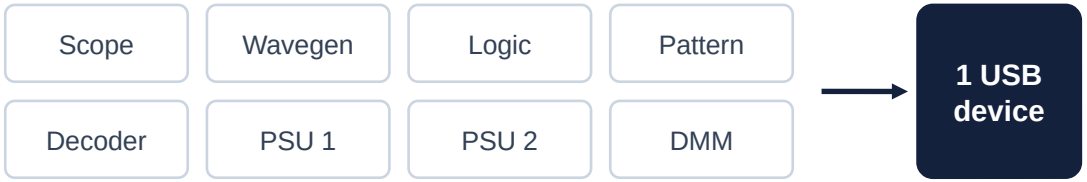
A complete electronics bench needs five to eight separate instruments.

Disconnected

Each instrument has its own interface; results are hard to combine and record.

Hard to interpret

Students can read a number but often not what it means, or what to fix.

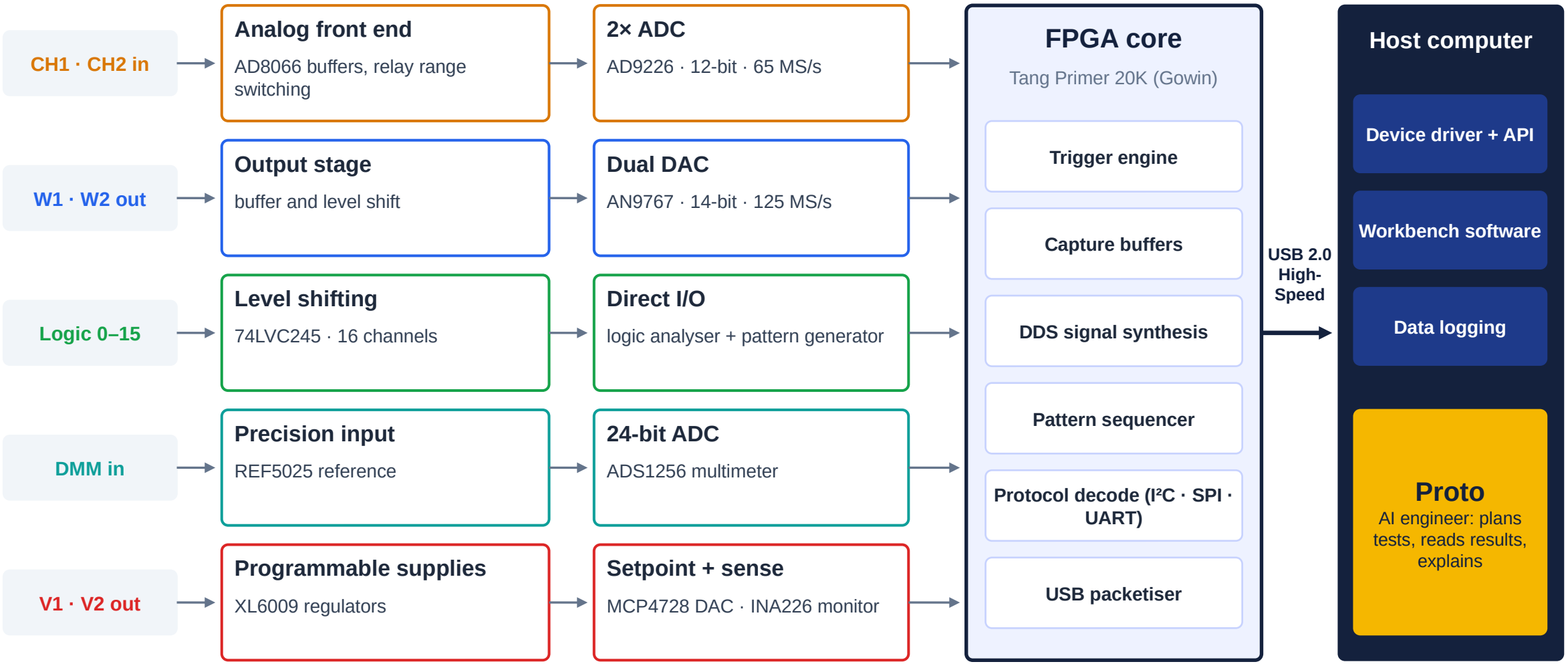


OBJECTIVES

1. **Integrate** eight instruments into one compact, USB-connected unit
2. **Design** an FPGA-based acquisition and signal-generation core
3. **Develop** a single software workbench for every instrument
4. **Embed Proto** an AI engineer that plans tests and explains results in plain language
5. **Keep it affordable** for student labs, with a low-cost production design

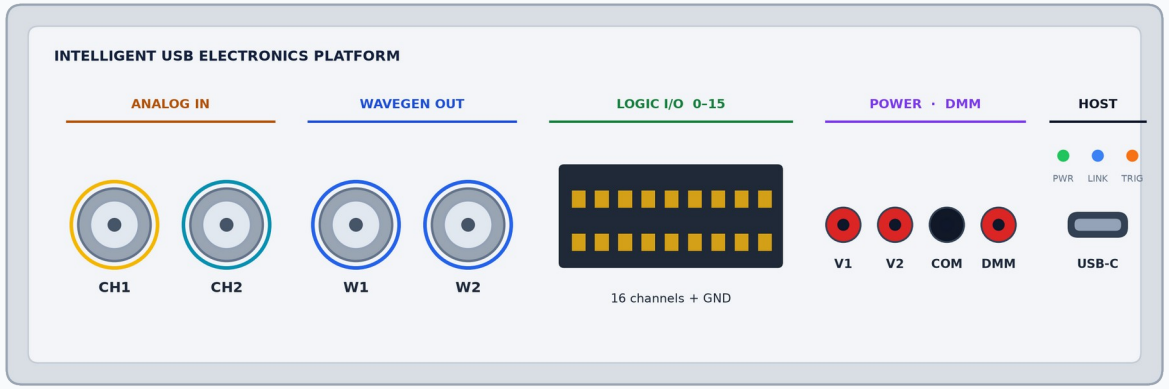
HARDWARE DESIGN

System architecture: one FPGA core, one USB link



HARDWARE DESIGN

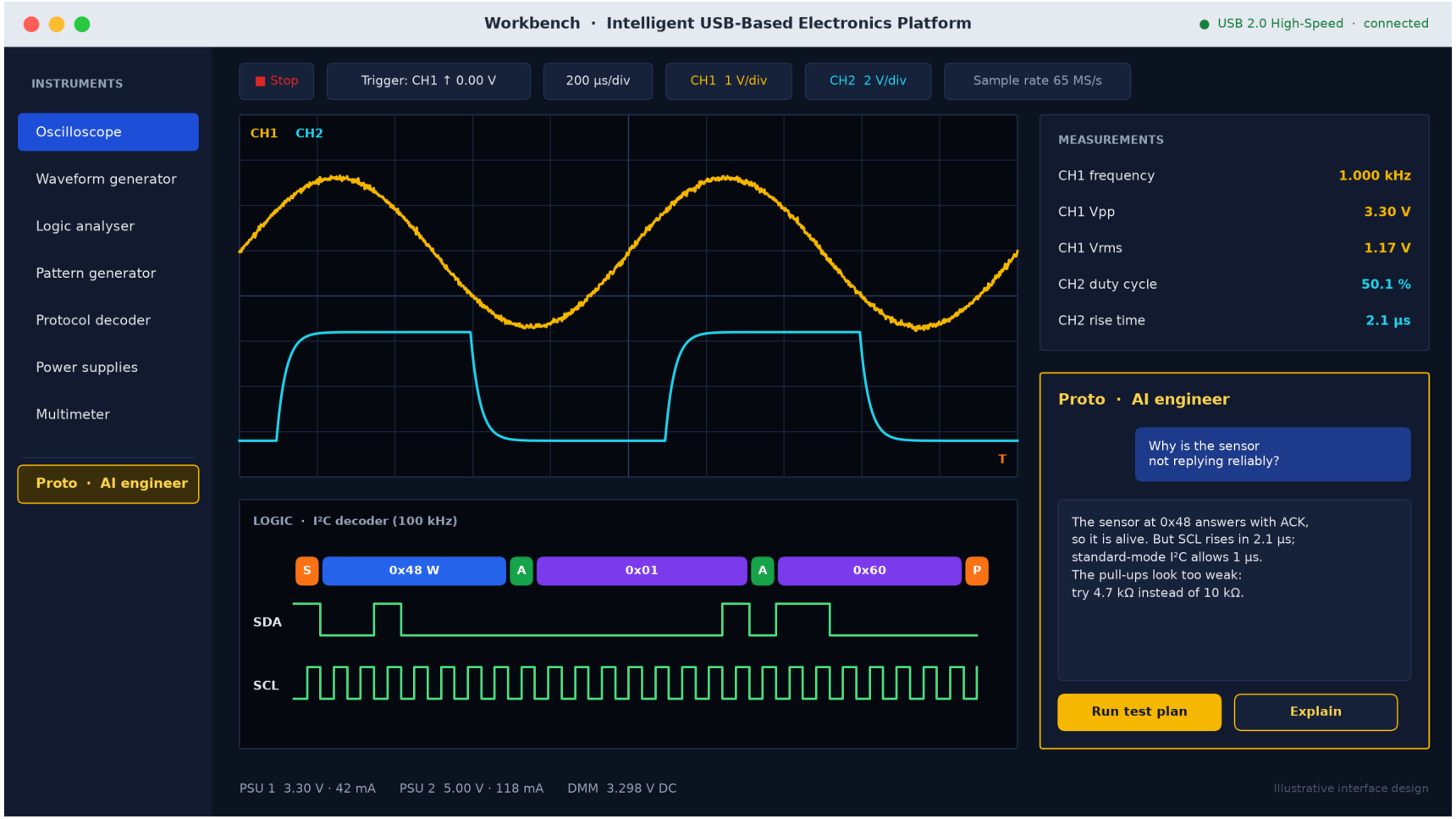
Instrument interface: the front panel and what each port does



Instrument	Port	Key specification	Hardware
Oscilloscope	CH1, CH2	2 channels · 12-bit · 65 MS/s	AD8066 front end + 2× AD9226
Waveform generator	W1, W2	2 channels · 14-bit · up to 125 MS/s	AN9767 dual DAC, FPGA DDS
Logic analyser	Logic 0–15	16 channels	74LVC245 + FPGA capture
Pattern generator	Logic 0–15	16 channels, programmable sequences	FPGA sequencer
Protocol decoder	Logic 0–15	I²C · SPI · UART	FPGA + host software
Programmable supplies	V1, V2, COM	2 outputs, voltage and current monitored	MCP4728 + XL6009 + INA226
Multimeter	DMM, COM	24-bit resolution	ADS1256 + REF5025

SOFTWARE DESIGN

Software interface: every instrument and Proto in one workbench



SOFTWARE STACK

Proto, AI engineer
plans tests, explains results

Workbench UI
scope, logic, decoder, supplies, DMM views

Device API
commands, streaming, logging

USB transport
USB 2.0 high-speed data link

FPGA gateware
trigger, capture, DDS, decode

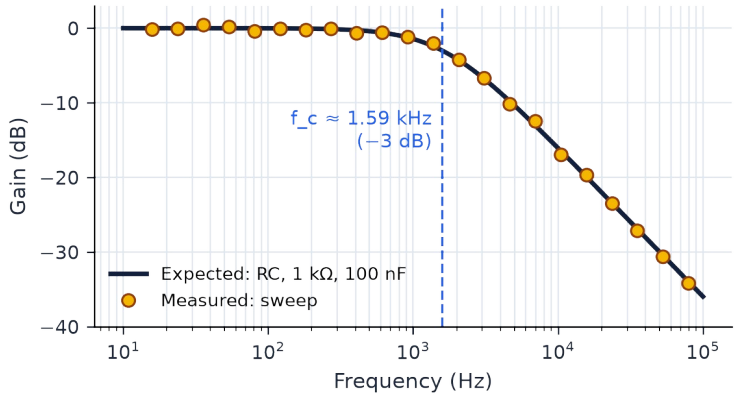
Top: what the user sees. Bottom: what runs on the hardware.

Proto, the AI engineer: from a measurement to a verdict

HOW PROTO RUNS A TEST

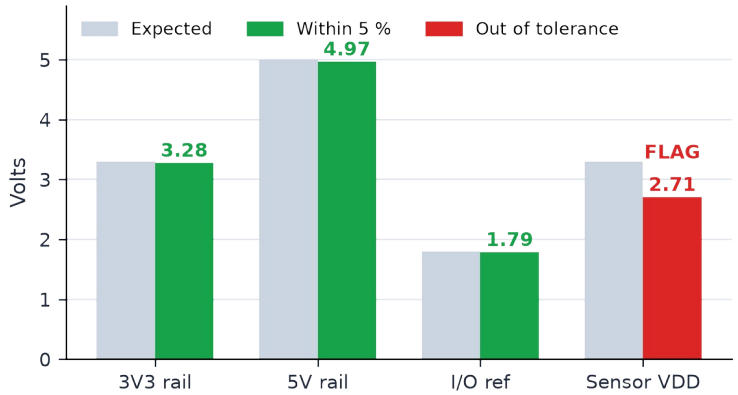
1. Connect the board
2. Power up, watch the current
3. Check every supply rail
4. Drive the inputs
5. Capture the outputs
6. Decode the buses
7. **Verdict in plain words**
8. **Suggest the fix**

Frequency response of an RC filter



Wavegen sweeps, scope measures, Proto compares with theory: f_c = 1/(2πRC) ≈ 1.59 kHz.

Supply-rail check with tolerance



DMM and supply monitors read each rail; anything outside ±5 % is flagged.

PROTO'S VERDICT

“The 3V3 and 5V rails are fine, but the sensor supply is 2.71 V, 18 % low. Check the regulator enable pin and the 10 µF output capacitor, then run the test again.”

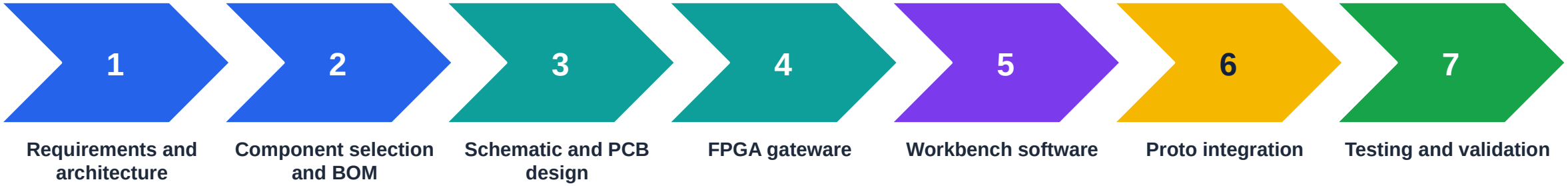
Graphs show illustrative data for the test method.

PLAN AND CONCLUSION

Methodology, outcomes and future scope

Thank you

DEVELOPMENT METHODOLOGY



EXPECTED OUTCOMES

- One USB unit replaces eight bench instruments
- Tests run as plans, not as manual steps
- Measurements explained in plain language by Proto
- Affordable enough for every student lab bench

FUTURE SCOPE

- A probe fixture that reaches every net of a board automatically
- More protocol decoders (CAN, I²S, 1-Wire)
- Remote lab access over the network
- Proto-generated lab reports from recorded measurements